

16 Apr '26

OS

Memory Management

Non-Continuous Memory - Paging

0 bit
1 bit

1 Byte = 8 bits

Word
→ 8 bits
→ 16 bits
→ 32 bits (common)

1 KB → 2^{10} bytes

1 MB → 2^{20} bytes

1 GB → 2^{30} bytes

1 TB → 2^{40} bytes

1 PB → 2^{50} bytes

~~1 X~~

Q 10 bit address → size of location 1 bytes

→ $2^{10} \times 1 \text{ bytes}$
1 KB

Q 44 bit address → size of location 2 bytes

→ $2^{44} \times 2 \text{ bytes}$
 $2^4 \text{ TB} \times 2 \text{ bytes}$
32 TB

Q 64 GB memory

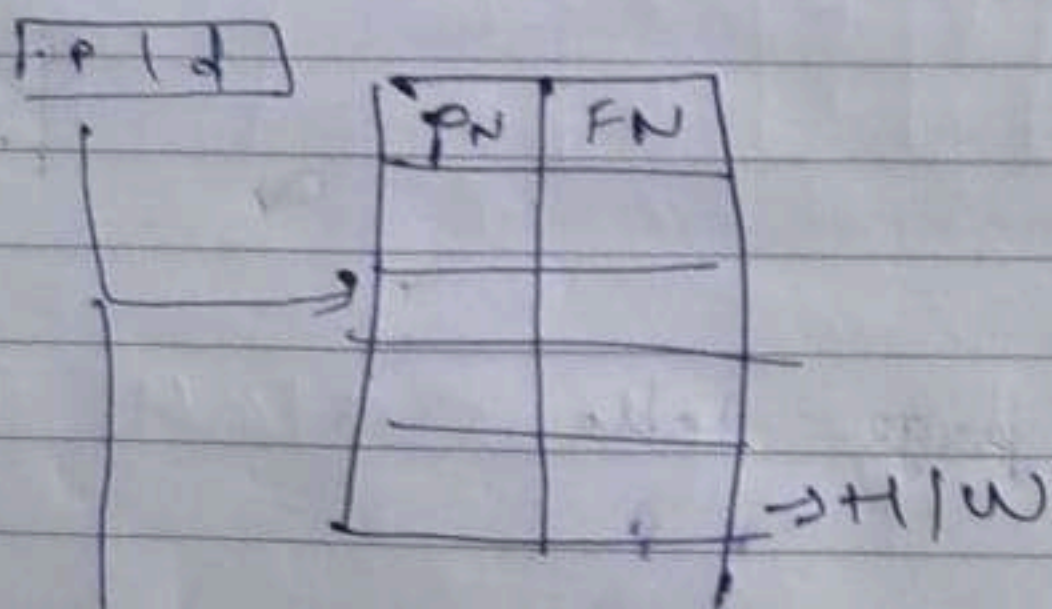
→ 32 bit address

~~$2^{32} \times 16 \text{ GB} \times 2^{30} = 16 \text{ bytes}$~~
 ~~4×2^{32}~~

Locality of Reference

- ↳ Spatial Locality - space
- ↳ Temporal Locality - time

Translation Lookaside Buffer $\rightarrow$ stores frame, page number so can access directly.



Page Table (if not found in TLB)

$$EMAT = H(TLB + MMAT) + (1 - H)(TLB + 2MMAT)$$

(with TLB)

$$EMAT = 2MMAT$$

$$MMAT = 400 \text{ microseconds}$$

$$TLBR = 50 \text{ microseconds}$$

$$\text{Hit ratio} = 90\%$$

$$EMAT = \frac{0.9}{100} (50 + 400) + 0.1 (50 + 800)$$

$$= 0.9 \times 450 + 0.1 \times 850$$

$$= 405 + 85$$

$$= 490$$

$$EMAT = 800$$

Ans

$$\frac{6}{10} \times \frac{5}{9} + \frac{4}{10} \times \frac{3}{9}$$
$$\Rightarrow \frac{30+12}{90} \Rightarrow \frac{42}{90} = \frac{1}{1.5}$$

Ans 2

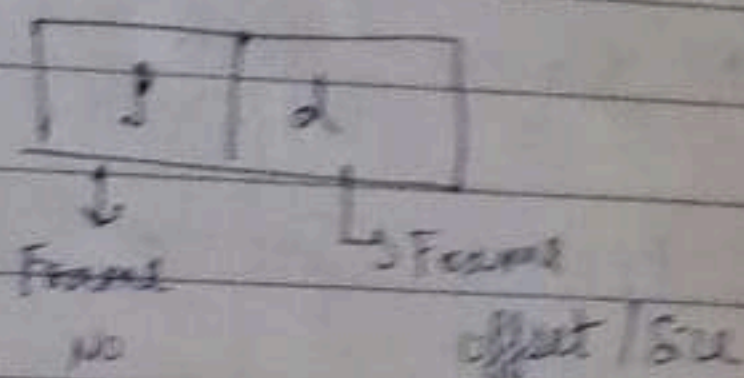
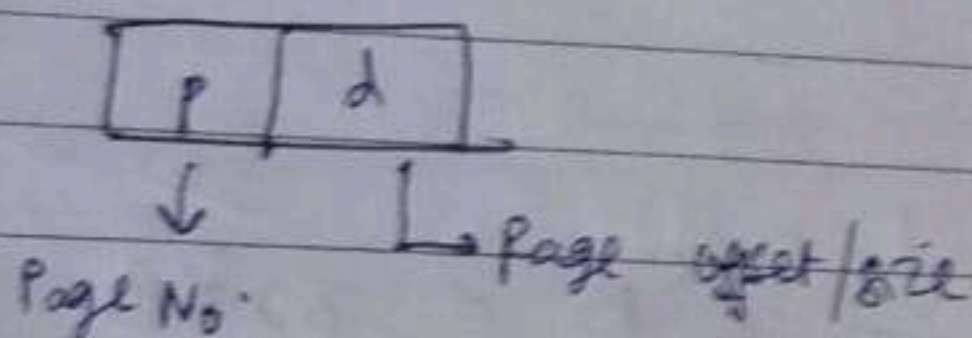
23rd Apr '26

OS

Paging / Non continuous memory allocation

Main Memory size $\rightarrow$ Physical Address space Frames

Logical Address space $\rightarrow$ Secondary Memory size Pages



Q

$$\text{No. of pages} = 2^{20}$$
$$\text{Page size} = 2B$$

$$\text{Size} \rightarrow 2^{20} \times 2B$$
$$\text{Memory} \quad 2MB$$

Q

$$\text{SM size} = 64GB$$
$$\text{Word addressable memory} = 4B$$

$$\text{No. of pages} = \frac{64GB}{4B} \rightarrow \frac{64 \times 2^{30}B}{4B}$$
$$= 16 \times 2^{30} = 2^{34}$$

Q

L.A = 32 bit

P.A = 16 bit

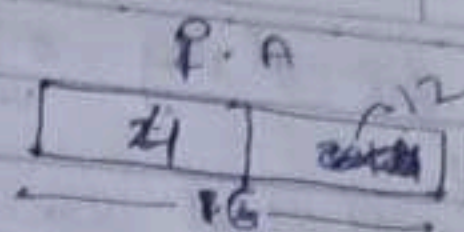
No. of frames = 4 bit

No. of pages = ~~30~~ bit

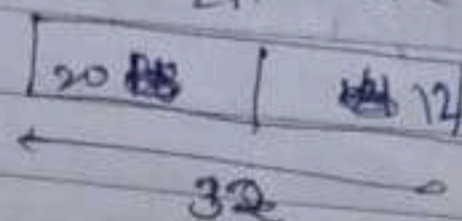
~~18~~ bit

20 bit

Page No.	YOUVA
Date	



LA Frame size = 2^{12}



22nd Apr '26

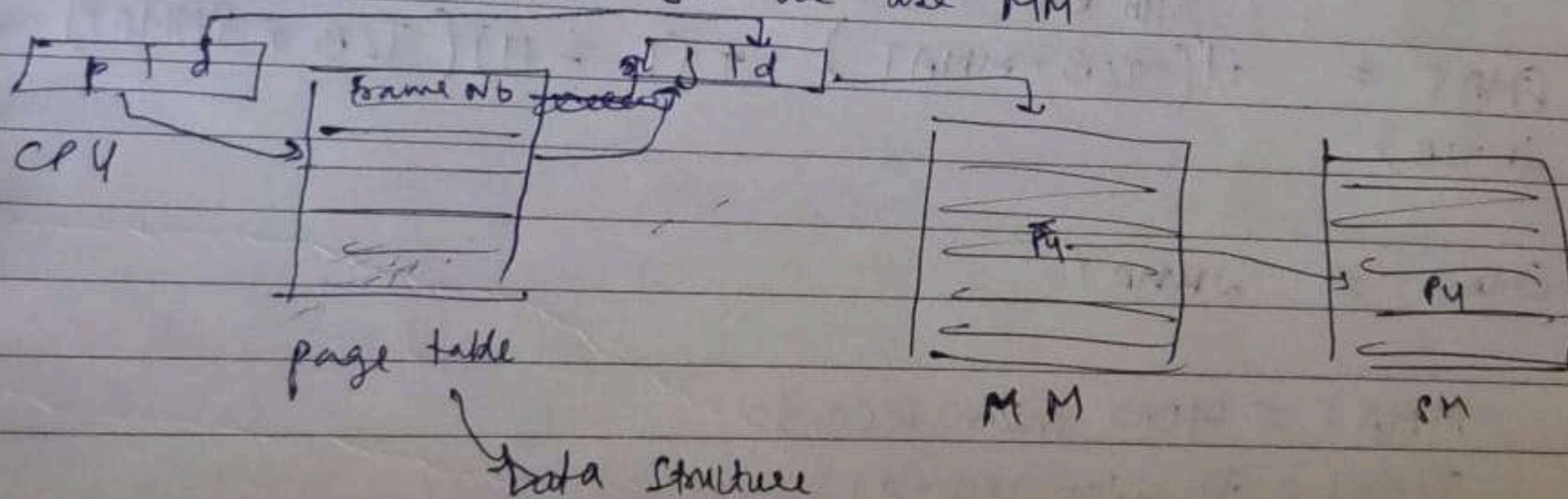
OS

Address Mapping

Every process has its own page table

Address Translation

To access SM we use MM



Effective Memory Access Time = $2 \times$ Main Memory Access Time

Size of Page Table = Number of processes
= Page Table Entry Size

Main memory size = No. of frames $\times$ frame size

$$[\text{Page size} = \text{Frame size}]$$

Q MM = 32 MB
 No. of frame = 2048

$$\text{Frame size} = \frac{32 \times 2^{20} \text{ B}}{2048 = 2^{11}} = 2^{14} \text{ B} = 16 \text{ KB}$$

Q MM = 256 MB
 SM = 512 MB
 No. of frames = 4096

$$\text{Frame size} = \frac{256 \times 2^{20}}{2^{12}} = 2^{16} \text{ B} = 32 \text{ KB}$$

$$\text{Page size} = 32 \text{ KB}$$

$$\text{No. of Pages} = \frac{2^9 \times 2^{30}}{2^{16}} = 2^{23} = 8388608$$

Q Logical
 Physical A = 64 Bit

$$\text{No. of Pages} = 64$$

$$\text{Page size} = 2^{58}$$

